

OOS Impact Analytics — Methodology Sheet

How the dashboard detects out-of-stock and demand-throttling events and values their revenue / contribution-margin impact. Worked example: **VV-VITA-231 — Myo-Inositol Complex Capsules**, analysed EU-wide (Pan-EU). Prepared for review.

1. Data sources

- **Amazon FBA Inventory Ledger** — real daily sellable warehouse balance and units shipped per SKU. The account runs **Pan-EU**, so stock is pooled: a SKU is physically out only when the whole EU balance hits zero.
- **Novadata margin export** — daily units, sales and contribution margin (CM3) per SKU and marketplace; supplies price and CM3 per unit for valuation.
- **Shopify catalog** — English product titles.

2. Two core measures

Demand rate (lambda). Trailing 90-day average units per calendar day, per SKU and marketplace. Daily sales follow an approximate Poisson process, so the chance of selling zero on a normal day is $P(0)=e^{-\lambda}$: ~37% at 1/day, ~5% at 3/day, ~0.7% at 5/day. A zero-sales day is therefore only treated as a stock-out when lambda is high enough that zero is a real anomaly (default threshold 3/day).

Reach (days-of-supply). EU sellable stock divided by trailing 28-day average units shipped — how many days of cover remain.

3. Daily classification (one label per SKU x marketplace x day)

Evaluated in priority order; the first match wins:

Priority	Label	Condition	Counts as
1	Physical OOS	EU sellable balance = 0	Lost (involuntary)
2	Critically low	Reach < 3 days	Lost (involuntary)
3	Cooling down	$3 \leq \text{reach} < 30$ days, still selling (units > 0), throttled (price $\geq +8\%$ and/or ad cut $\geq 50\%$)	Miss (voluntary)
4	Demand gap (EU)	Units = 0 on a day enclosed by sales, $\lambda \geq 3/\text{day}$ (incl. throttles that hit 0)	Lost (involuntary)

Guards: the SKU must already be selling (ignores pre-launch); a zero-run must be enclosed by later sales or be within the last 21 days (ignores discontinued tails). Cooling-down separates deliberate throttling — which otherwise looks like a demand drop — from genuine stock-outs, so the two are never double-counted.

4. Impact valuation

On any flagged day: **missed units = $\max(\lambda - \text{actual units}, 0)$** (the expected demand we didn't capture). Valued at the SKU's trailing average selling price and CM3 per unit:

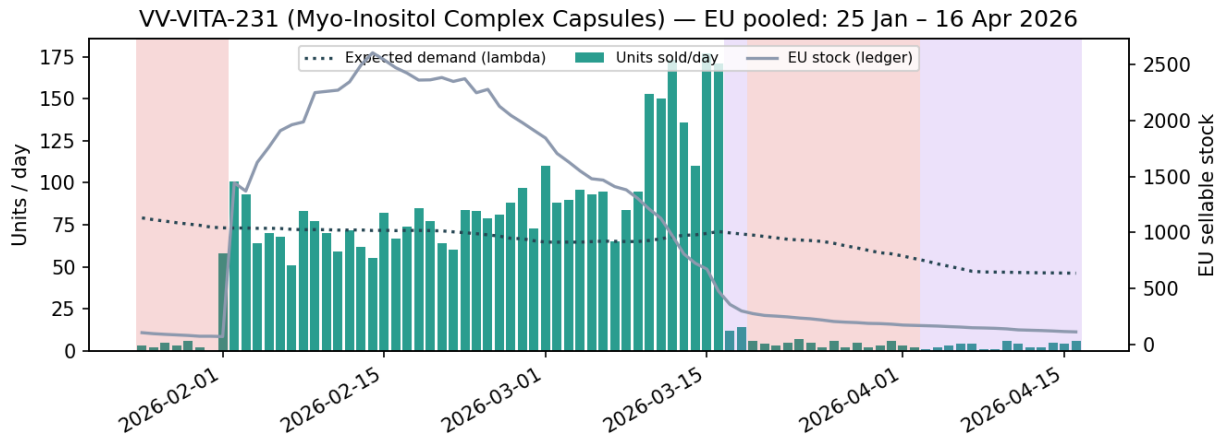
Lost / Miss revenue = missed units x avg price

Lost / Miss CM3 = missed units x avg CM3 per unit (the true P&L impact).

'Lost' = involuntary (OOS); 'Miss' = voluntary (cooling down). Consecutive flagged days are grouped into discrete events.

5. Worked example

VV-VITA-231 — Myo-Inositol Complex Capsules, EU-wide (Pan-EU pooled). Full year: **38 OOS days** (lost €67.971 revenue / €21.260 CM3) and **27 cooling-down days** (miss €34.917 revenue / €8.808 CM3).



Highlighted window 25 Jan 2026 – 16 Apr 2026: EU stock ranged from ~2,607 down to ~70 units and reach bottomed at ~2 day(s). In this window the model flagged **23 OOS day(s)** (red/orange shading) and **16 cooling-down day(s)** (purple) — the SKU was throttled (price up / ad spend cut) while stock was tight, then went into genuine stock-out as reach collapsed. Note the balance rarely hits literally zero, so the **reach < 3** rule, not a balance-of-zero rule, is what catches the hard stock-out. Stock was replenished on **02 Feb** (EU balance jumps in the chart), after which sales recover and the flags clear — the table's last rows show those first restock days.

What "cooling down" means here. On the purple days the SKU kept selling (units > 0) but demand was deliberately damped to stretch the remaining stock to the next shipment. **Before cooling down** it sold at ~€28.62 with ~€409/day of ad spend; **during the throttle** the price was lifted to ~€30.94 and ad spend cut to ~€25/day. Sales fell to about 4/day versus the expected ~68/day, and that gap is booked as a **voluntary miss** — a pricing / advertising choice to protect availability and the listing's ranking — not a lost stock-out. Crucially the SKU keeps selling: the moment a throttle pushes sales to zero it is no longer cooling down and counts as OOS (lost). Cooling down trades a little volume now (the miss) to avoid the larger ranking and lost-sales hit of a full stock-out.

Representative days (Lost/Miss shown as CM3 €):

Date	Units	λ	EU stk	Reach	Price	Category	Lost €	Miss €
01-25	3	79	105	2	31.14	Critically low (<3d)	711	
01-26	2	78	96	2	31.07	Critically low (<3d)	709	
01-27	5	77	90	2	31.07	Critically low (<3d)	672	
01-28	3	76	85	2	31.05	Critically low (<3d)	681	
01-29	6	75	80	2	31.05	Critically low (<3d)	644	
01-30	2	75	73	2	31.14	Critically low (<3d)	675	
02-01	58	73	70	2	28.73	Critically low (<3d)	140	
02-02	101	73	1445	55	28.63	—		
02-03	93	73	1371	53	28.72	—		
02-04	64	73	1628	74	28.70	—		
02-05	70	73	1763	84	28.73	—		
03-17	12	70	357	4	29.08	Cooling down		411
03-18	14	70	299	3	29.82	Cooling down		390
04-03	1	53	167	3	30.98	Cooling down		348
04-04	2	52	164	3	30.93	Cooling down		328
04-05	3	50	159	3	30.93	Cooling down		312

6. Caveats

Impact figures are estimates (assume the SKU would have sold at its recent rate). Ad-spend-cut detection only applies from when Novadata began reporting Advertising Costs (~Feb 2026); the price-hike lever spans the full year. Reach is EU-pooled, so reach-based rules need ledger coverage for the SKU. All thresholds (3/day, reach 3 & 30, +8%, -50%) are tunable in the dashboard.